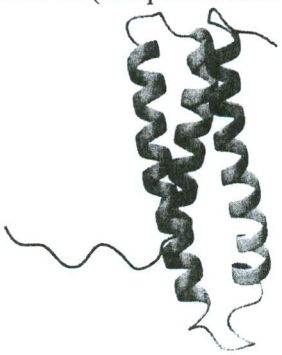
Biochemistry Cume 4/24/15

1. (20 pts) Sketch a titration curve for the amino acid Ala and indicate the pKa values for all titratable groups. Label the axes appropriately. Also indicate the pH at which this amino acid has no net charge.

2. (10 pts) Estimate the maximum dimension of the helical protein that is shown below. (Helpful hints: 5.4 Å rise/turn, 3.6 residues/turn).



3. (70 pts) Choosing from the list below, name the molecules that are described. Answers can be used more than once, and some blanks will require more than one answer. Choose the best answer(s) for each description.

List: Asn, Gln, Asp, Gly, Ile Arg, Met, Pro, Ser, Phe, Val, Thr

A. A pair can form a salt bridge (electrostatic attraction) using their side-chains _____
B. The geometry of this imino acid is such that it does not fit into the alpha-helix _____
C. Non-polar, more atoms than Val, and find them in the center of a protein _____
D. Two chiral centers _____
E. The largest _____
F. The smallest _____
G. Electrically neutral polar side chains _____
H. Two residues that are most frequently found in a triple-helical collagen _____
I. An essential sulfur-containing monomer _____
J. Beta-branched _____
K. Closely resembling (isosteric to) Cys _____
L. Hydroxyl group _____
M. Amide group _____
N. Sodium salt of this food additive is commonly known as MSG _____
O. Achiral _____
P. Isomer (same molecular formula) of Leu _____